

Chi¹
1 SunMintaka
4,945 SunsAlnitak
8,940 Suns

ORION THE HUNTER

FORMED FROM AN EASILY RECOGNIZABLE PATTERN OF STARS, ORION IS FAMILIAR TO MOST SKYWATCHERS. IT CONTAINS SEVERAL BRIGHT STARS AND THE ORION NEBULA, ONE OF THE MOST BEAUTIFUL SIGHTS IN THE NIGHT SKY.

Orion is an ancient constellation that represents a hunter or warrior in Greek myth. Orion was the son of Poseidon, the god of the sea, and was a hunter of great prowess. Despite his hunting skill, he was killed by a mere scorpion, possibly in retribution for his boastfulness. In the sky, the scorpion is depicted by the constellation Scorpius, and as Orion sets below the horizon, Scorpius rises and pursues him across the sky. Close to Orion's heels are Canis Major and Canis Minor, representing his hunting dogs.

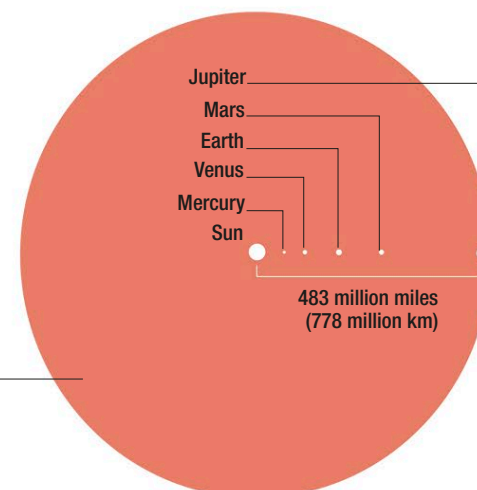
The two brightest stars in Orion provide a striking colour contrast: the red supergiant Betelgeuse marks the hunter's shoulder, while the blue supergiant Rigel is positioned in one of his feet. Many of the constellation's highlights are near the line of stars that represents Orion's Belt. The belt is easy to find because it is made up of three equally spaced bright stars—Alnitak, Alnilam, and Mintaka—which form an almost perfectly straight line. Just below the belt is a complex of stars and nebulae that represent the hunter's sword. This area includes a vast area of star formation called the Orion Nebula (M42), the largest and closest nebula of its kind in the night sky. Other nebulae lie nearby, including the Horsehead Nebula, a dark nebula silhouetted against the bright emission nebula, known as IC 434.

When Betelgeuse explodes, it will release more energy in an instant than the Sun will produce in its lifetime

▷ Betelgeuse

The supergiant star Betelgeuse is more than 500 times bigger than the Sun. If it was placed at the center of our Solar System, it would engulf the Sun and all the planets out as far as Jupiter. Betelgeuse is both relatively young and highly unstable, varying erratically in brightness. At some point in the next million years, it will probably explode in a supernova.

Betelgeuse
Radius 510 million
miles (820 million km)

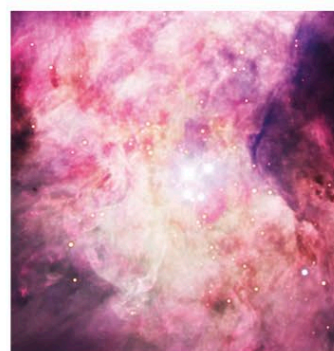


▷ M42

Better known as the Orion Nebula, this star-forming region is about 24 light-years across. It is shown here as bright pink because radiation from hot young stars causes hydrogen gas to glow pink. The nebula is embedded in a much larger dark cloud; dust in this cloud is coloured dull pink.

▽ The Trapezium

The stars forming in the Orion Nebula include a group called the Trapezium. As well as the four stars seen here, the system includes two fainter members.



▷ Star distances

At just under 500 light-years away, Betelgeuse is the closer of Orion's two brightest stars. Rigel is much more distant—about 860 light-years away—but most of the time, Rigel looks brighter than Betelgeuse because it emits far more light. The three stars that form the line of Orion's Belt are widely spread out in space, with Alnilam being the most distant. In fact, Alnilam is the farthest of all Orion's main pattern stars, at nearly 2,000 light-years away. The nearest pattern star is Pi³ Orionis, which is only 26 light-years from Earth

